A Weil-positivity lock detects and counts genuine off-line zeros of an Epstein zeta function

Preprint, version 1.0.1

Mehmet Akif Oltulu
Maovera AI Research (independent), Istanbul, Türkiye
ORCID: 0009-0003-0928-0854

18 October 2026

Abstract

Weil's explicit formula turns the location of the zeros of a zeta function into the sign of a quadratic form, and the resolvent lock of [3] reads that sign, together with the exact count of negative directions, from the real zeros of a single polynomial. We build the truncated Weil form of a two-dimensional Epstein zeta function directly from its own data—the gamma factor, one constant and a finite set of *generalised primes* obtained from the logarithmic derivative of its Dirichlet series by an exact recursion—so that the construction uses the function's own data only. The Epstein family has no Euler product and, for suitable lattices, genuine zeros off the critical line [5]. On the lattice $D = 7$, whose zeros all lie on the line, the lock is closed ($\kappa = 0$). On the lattice $\Delta = 0.800942\ldots$ the lock opens with $\kappa = 4$, and the index law splits the count as $4 = 2 + 2 + 0$. Subtracting the zero-side contributions of the three off-line quadruples inside the band closes the lock again ($\kappa = 0$); subtracting any single one leaves it open. Along the way the explicit-formula identity, evaluated from function values, located two off-line zeros beyond the published one, at heights 17.21 and 23.06 (the latter with real part 1.326). All numbers were reproduced on two machines.

Scope. Computed (100 digits, two machines, identical to the printed digits): the Epstein Weil form, its eigenvalue counts and the lock readings of Table 3. Computer-verified (interval arithmetic): the lock's zero-count certificates. Verified against independent methods: the construction reproduces the calibrated ζ machine of [3] to $1.5\cdot10^{-58}$ and the explicit-formula identity to $3.9\cdot10^{-32}$.

Keywords: Weil explicit formula, Weil positivity, Epstein zeta function, off-critical zeros, generalised primes, Pontryagin index, resolvent lock, Chowla–Selberg formula

1 Introduction

Weil's criterion [1,2] states that the Riemann hypothesis is equivalent to the positivity of a Hermitian form built from the explicit formula. Criteria of this kind are usually examined on ζ itself, where every computed zero lies on the line; recent numerical realisations test them on synthetically injected off-line zeros [8]. The Epstein zeta functions of binary quadratic forms supply a natural family with a functional equation of the same shape, no Euler product, and genuine off-critical zeros [5,7]. This note applies the finite-window Weil form and the resolvent lock of [3] to two members of that family whose zero sets are known independently: one with all zeros on the line and one with off-line zeros. The lock is closed on the first, opens on the second, and the opening is accounted for zero by zero.

2 The Weil form from function values

Let $D > 0$ and $Z(s) = \sum'_{(j,k)\in\mathbb{Z}^2}(j^2+Dk^2)^{-s}$. The completed function $\Lambda(s) = (\sqrt{D}/\pi)^s\Gamma(s)Z(s)$ satisfies $\Lambda(s) = \Lambda(1-s)$ and has simple poles at $s = 0, 1$. For φ even-symmetric in the sense

$\varphi = f * \tilde{f}$ with f supported in $[-L/2, L/2]$ and $F(s) = \int \varphi(u) e^{(s-1/2)u}\, du$, contour integration gives the explicit formula

$$\sum_{\rho} F(\rho) = F(0)+F(1)+2\kappa\varphi(0)+\int_0^\infty \Big[\frac{2e^{-x}\varphi(0)}{x} - \frac{\varphi(x)+\varphi(-x)}{2\sinh(x/2)}\Big]dx + \sum_{w} C_w\, w^{-1/2}\big(\varphi(\log w)+\varphi(-\log w)\big), \tag{1}$$

with $\kappa = \log(\sqrt{D}/\pi)$ and $Z'/Z(s) = \sum_w C_w w^{-s}$. The coefficients C_w (*generalised primes*) follow exactly from $Z' = (Z'/Z)\, Z$: ordering the multiplicative semigroup generated by the values $j^2 + Dk^2 > 1$,

$$r_1\, C_w = -r(w)\log w - \sum_{uv=w,\ v\neq 1} C_u\, r(v), \tag{2}$$

where $r(v)$ counts representations and $r_1 = r(1)$. Keys are kept as integer polynomials in D, so the recursion is exact for every D. Because φ is supported in $[-L, L]$, only $w < e^L$ enter (1); for ζ the same recursion returns $C_n = -\Lambda(n)$ and (1) becomes Weil's formula with $\Gamma(s/2)$ in place of $\Gamma(s)$. In the Fourier basis $e_m = (-1)^m e^{2\pi i m u/L}/\sqrt{L}$, $|m| \le M$, every entry $Q_{mn} = W(e_m * \tilde{e}_n)$ is a closed form: a rank-two pole term $2\,\mathrm{Re}(v_m v_n)$ with $v_m = 2\sinh(L/4)/(\sqrt{L}(\frac{1}{2} + 2\pi i m/L))$, one-dimensional archimedean integrals, and the shift kernels

$$P_{mn}(x) = \frac{\sin(2\pi n x/L) - \sin(2\pi m x/L)}{\pi(m-n)}\ (m \neq n), \qquad P_{nn}(x) = 2\Big(1 - \frac{x}{L}\Big)\cos\frac{2\pi n x}{L}. \tag{3}$$

The zero side of (1) gives, for a zero ρ with $t = (\rho - \frac{1}{2})/i$, the contribution $\hat{e}_m(t)\,\overline{\hat{e}_n(t)}$; on the line it is positive semidefinite, and an off-line quadruple contributes a form of signature $(2,2)$. The lock of [3, Theorem 6] then reads the number κ of negative eigenvalues of Q from the polynomial of $x = Q^{-1}e_0$: $\kappa = N_c(x) + d(x) + [x_0 < 0]$.

3 The test family

Two rectangular lattices are used. *Clean:* $D = 7$ (equivalently $\Delta^2 = 1/7$), whose zeta function factors through ζ and an L-function; all its computed zeros lie on the line. *Dirty:* $\Delta = 0.800942429865856$, $D = \Delta^{-2}$, near an edge point of [5], with a known off-line zero at height 9.209. The function itself is evaluated by the Chowla–Selberg expansion [6]. Window $L = \log 13$, $M = 10$ and 12 (bands 24.5 and 29.4), 100 digits.

4 Results

Table 1 lists the off-line zeros of the dirty lattice, refined at 60 digits. The first is the published one [5]; the second was located from the residual of the explicit-formula identity ($1.85 \cdot 10^{-5}$, matching its quadruple's contribution $1.853 \cdot 10^{-5}$); the third and fourth from the argument-principle count, which gives 8 zeros up to height 20.5 and 15 up to 30.5, in agreement with the zeros found. Table 2 checks (1) with the Gaussian $\hat{\varphi}(t) = e^{-t^2/2\tau^2}$, $\tau = 3.5$, against the zeros

Table 1: Off-line zeros $\rho = \frac{1}{2} + a + i\gamma$ of the dirty lattice (60 digits). Last two columns: $|\Lambda(\rho)|$ and $|\Lambda(\rho + 0.01)|$.

γ	a	$\lvert\Lambda(\rho)\rvert$	$\lvert\Lambda(\rho+0.01)\rvert$
9.20895633677026	0.242656369701712	$8.8 \cdot 10^{-67}$	$1.0 \cdot 10^{-8}$
17.2135107028407	0.431059253114527	$1.2 \cdot 10^{-68}$	$1.8 \cdot 10^{-13}$
23.06306483933	0.826378055331069	$2.0 \cdot 10^{-68}$	$3.8 \cdot 10^{-17}$
30.9462330328128	0.145571120365216	$1.1 \cdot 10^{-69}$	$3.7 \cdot 10^{-23}$

Table 2: Explicit-formula identity: function side minus zero side.

Clean lattice, critical zeros below 41	$3.89 \cdot 10^{-32}$
Dirty lattice, with the four quadruples	$-3.42 \cdot 10^{-23}$
Dirty lattice, critical zeros only	0.1238
Control: ζ kernel $\Gamma(s/2)$ in place of $\Gamma(s)$	-2.43

themselves. Table 3 is the main result. The clean form is positive and the lock is closed. The dirty form has four negative eigenvalues and the lock opens with the same count. Removing the zero-side contributions of the three in-band quadruples (heights 9.21, 17.21, 23.06) leaves a positive form, $\lambda_0 \approx 10^{-15}$, and the lock closes; removing only the first leaves $\kappa = 4$. The structure residual of every matrix is at the working precision, so the theorems of [3] apply to each row.

Table 3: Resolvent lock on the Epstein Weil form ($L = \log 13$). Index law: $\kappa = N_c + d + [x_0 < 0]$.

M	form	λ_0	κ	lock	$N_c + d + [x_0 < 0]$
10	clean, $D = 7$	0.60537	0	closed	$0 + 0 + 0$
10	dirty, $\Delta = 0.8009$	-0.43720	4	open	$2 + 2 + 0$
10	dirty $-$ 3 in-band quadruples	$2.6215 \cdot 10^{-15}$	0	closed	$0 + 0 + 0$
10	dirty $-$ all 4 quadruples	$2.5535 \cdot 10^{-15}$	0	closed	$0 + 0 + 0$
10	dirty $-$ height-9.21 quadruple only	-0.35592	4	open	$4 + 0 + 0$
12	clean, $D = 7$	0.59692	0	closed	$0 + 0 + 0$
12	dirty, $\Delta = 0.8009$	-0.47657	4	open	$2 + 2 + 0$
12	dirty $-$ 3 in-band quadruples	$1.0351 \cdot 10^{-15}$	0	closed	$0 + 0 + 0$
12	dirty $-$ all 4 quadruples	$1.3844 \cdot 10^{-16}$	0	closed	$0 + 0 + 0$
12	dirty $-$ height-9.21 quadruple only	-0.39556	4	open	$4 + 0 + 0$

5 Controls and pre-registration

Every check prints a correct input that passes next to a corrupted input that fails (Table 4). Pre-registered before the live run and confirmed on both machines: the identity bounds (P13), $\kappa = 0$ and a closed lock on the clean lattice (P14), $\kappa = 4$ and an open lock on the dirty lattice (P15), a closed lock after removing the in-band quadruples (P16), and an open lock after removing a single quadruple (negative control). Two predictions made at the sandbox stage were overturned by measurement and are recorded here: at this window the ground-state zeros of the Epstein form follow their own interlacing pattern (the zeros are twice as dense as those of ζ, the spectrum is full and $\lambda_0 \approx 0.6$); and removing two quadruples left $\kappa = 2$, which led to the third, in-band quadruple at height 23.06.

Table 4: Self-tests of the construction (each with a corrupted input).

Generalised primes	$D = 1, 2$: exact $-\Lambda(n)(1 + \chi(n))$, error $6.0 \cdot 10^{-40}$	unit coefficient altered: $1.2 \cdot 10^{2}$
Chowla–Selberg	$Z(2; 1) - 4\zeta(2)G = 0$	wrong D: 2.5
Weil form, two routes	closed form vs. calibrated machine (ζ): $8.6 \cdot 10^{-31}$	sign of constant reversed: 2.3

6 Contributions of this note

1. *A Weil form without an Euler product.* The explicit formula needs only the functional equation and the Dirichlet series; the recursion (2) turns the latter into a finite, exact list of generalised primes for any window. This makes the positivity test available for every Epstein zeta function, and the construction is calibrated against the ζ machine of [3] to $1.5 \cdot 10^{-58}$.
2. *Attribution by removal.* Subtracting the zero-side contribution of selected zeros turns the lock into an accounting device: it identifies which zeros produce the negative directions, here exactly the in-band off-line quadruples.
3. *The identity as a zero finder.* The residual of (1), computed from function values, measures the contribution of zeros missing from a list; it located the quadruple at height 17.21 before any search.

7 Scope and continuation

To our knowledge this is the first application of a Weil-positivity form, built from the values of a zeta function without Euler product, to a family with genuine off-critical zeros, with an exact count of the negative directions they produce (literature search of 18 October 2026; nearest work: synthetic injection for ζ [8] and Sylvester-inertia arguments for hypothetical off-line pairs of ζ [9]). The continuation, subject of the next note: interval certificates for the eigenvalue signs, a sweep of Δ through the edge points of [5], and longer windows where the silent regime of [3] appears.

8 Data and reproduction

Tool `icice_sertifika_v1_4.py` (MD5 `e475b01a3b966f4f04304d20b0217047`), single file; run `--epstein --Mler 10,12 --dps 100`; live log of 18 October 2026 (534 s). The sandbox run and the live run agree in every printed digit.

References

[1] A. Weil, *Sur les "formules explicites" de la théorie des nombres premiers*, Comm. Sém. Math. Univ. Lund (1952), 252–265.

[2] E. Bombieri, *Remarks on Weil's quadratic functional in the theory of prime numbers, I*, Rend. Mat. Acc. Lincei (9) **11** (2000), 183–233.

[3] M. A. Oltulu, *Odd-sector, Pontryagin-space and index-law extensions of the Connes–van Suijlekom real-zeros theorem*, Zenodo, doi:10.5281/zenodo.23459270 (2026).

[4] A. Connes, W. D. van Suijlekom, *Quadratic forms, real zeros and echoes of the spectral action*, arXiv:2511.23257 (2025).

[5] L. Bétermin, L. Šamaj, I. Travěnec, *Interplay between critical and off-critical zeros of two-dimensional Epstein zeta functions*, arXiv:2110.09368.

[6] S. Chowla, A. Selberg, *On Epstein's zeta-function*, J. reine angew. Math. **227** (1967), 86–110.

[7] H. Davenport, H. Heilbronn, *On the zeros of certain Dirichlet series*, J. London Math. Soc. **11** (1936), 181–185, 307–312.

[8] T. Kim et al., *A numerical realization of Suzuki's Weil-quadratic-form operator*, arXiv:2607.24830 (2026).

[9] *More than two thirds of the zeta zeros are simple and on the critical line*, arXiv:2608.13637 (2026).

[10] P. Epstein, *Zur Theorie allgemeiner Zetafunctionen*, Math. Ann. **56** (1903), 615–644.